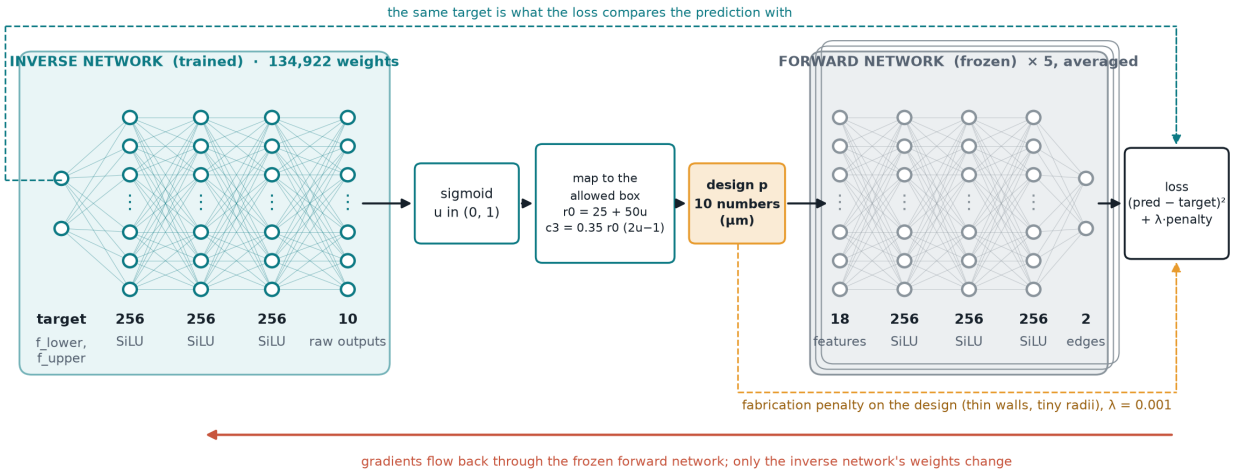
The networks and why they look like this

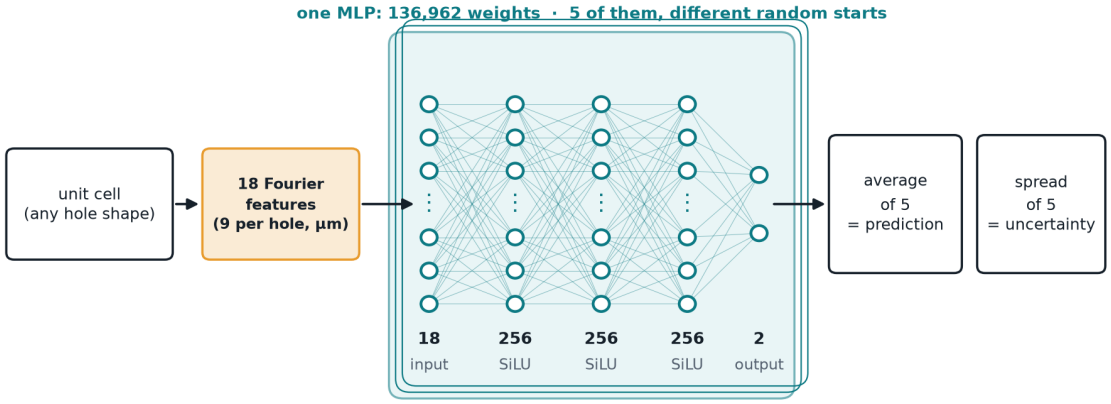
Diagrams of every network, the reason behind each setting, and the architecture test that backs the choices up. The test trained on the training split and scored on the validation split; the test split behind the reported results was never used.

1. The tandem network



The inverse network (left, trained) proposes a design; the frozen forward ensemble (right) predicts its band edges; the loss compares them with the target. Only the inverse network learns.

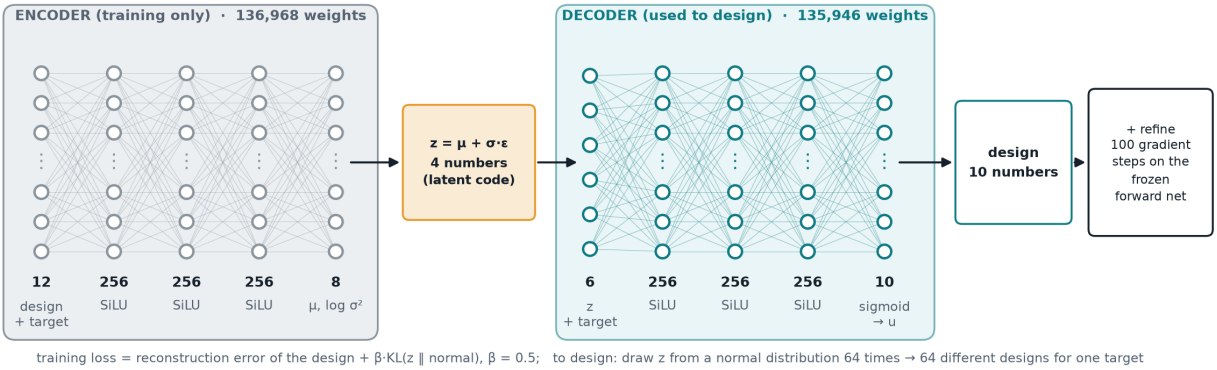
2. The forward ensemble



inputs and outputs are standardised (mean 0, spread 1) with training-set statistics; loss = mean squared error; AdamW, one-cycle schedule, 400 epochs, batch 128

Five identical networks from different random starts; their average is the prediction and their spread the uncertainty.

3. The conditional VAE (+ refine)



The encoder is only needed for training; to design, random codes z go through the decoder, and each design is polished by gradient steps.

Every setting and why

Part	Setting	Value	Why
Forward network	Inputs	18 Fourier numbers	9 per hole (harmonics 0, 3, 6, 9, 12): describes circles, triangles and freeform shapes in one format, so one network serves every family.
Forward network	Hidden layers x width	3 x 256	Validation test: 1 layer 0.61 GHz, 3 layers 0.21 GHz; larger sizes (best 4 x 512) gain 0.05 GHz, which changes no conclusion; an easy upgrade later.
Forward network	Activation	SiLU	Validation test: GELU 0.20, SiLU 0.21, Tanh 0.38, ReLU 0.57 GHz; smooth, so clean gradients for inverse design.
Forward network	Ensemble	5 networks	Validation test: 1 network 0.19 GHz, 5 networks 0.17 GHz; the spread flags risky predictions; 10 would double the cost for little gain.
Forward network	Training	400 epochs, batch 128, AdamW 2e-3, one-cycle	Validation test: 100: 0.39, 200: 0.30, 400: 0.21, 800: 0.15 GHz; 800 would be better: 400 was a speed trade-off (an easy upgrade).
Forward network	Loss	mean squared error on standardised edges	Smooth gradients; standardising makes both edges count equally. Accuracy is still reported as MAE in GHz.
Tandem	Inverse network	2 -> 256 x 3 -> 10	Validation test (share of targets with a fabricable design within 1 GHz): 2 x 128: 99.5 %, 3 x 256: 100 %, 4 x 256: 100 %, 3 x 512: 100 %.
Tandem	Output squashing	sigmoid + box mapping	Every output is a valid shape inside the sampled box, so the network cannot propose r0 = 200 um.
Tandem	Fabrication penalty	lambda = 0.001	Validation test: λ 0 -> 90.5 % usable, λ 0.001 -> 100 % usable, λ 0.01 -> 100 % usable.
Tandem	Training targets	band edges with gap > 0.5 GHz	Only real gaps are useful targets; no shapes are needed, which avoids the averaging trap.
CVAE	Latent code	4 numbers	Validation test: 2 -> 4.2 designs, 4 -> 4.6 designs, 8 -> 6.9 designs within 1 GHz per target (before refining); a larger code would help, refine makes up for most of it.
CVAE	beta	0.5	Validation test: β 0.1 -> 3.7 designs, β 0.5 -> 4.6 designs, β 1 -> 4.9 designs (before refining); trades accuracy of samples against variety.
CVAE	Samples per target	64	Enough to find about 8 distinct valid designs after refining, while design still takes only 0.2 s.
Search	Multi-start	64 starts x 300 Adam steps	Many random starts so that some avoid poor local minima; 300 steps converge.
Search	Refine	100 steps from CVAE samples	CVAE samples already start close, so 100 steps suffice: 3x fewer than from random starts.
Correction	Model	Gaussian process, ARD RBF + white noise	Smooth, small difference; GP needs no architecture tuning and gives an uncertainty. Length scales fitted on 700 cells then fixed so thousands of cells stay fast.

The architecture test

Depth and width

I tested it on the validation split (single networks, 3 random seeds each, scored in GHz, mean of both edges). With 256 neurons per layer: 1 layer 0.61, 2 layers 0.28, 3 layers 0.21, 4 layers 0.18, 5 layers 0.19 GHz. Going from 1 to 3 layers cuts the error by 66 %. The best network in the test was 4 x 512 at 0.16 GHz, 0.05 GHz better than 3

x 256 (0.21 GHz) with 5.8x the weights. 3 x 256 was fixed as a standard size before this test, and all reported results use it. The gain would not change any conclusion: the forward error (0.13 / 0.17 GHz for the ensemble) is already far below the 1 GHz design criterion and below the PWE-COMSOL difference, which dominates the final error. Using a larger network is an easy, free improvement for Semester IV.

Activation

Tested at 3 x 256 on the validation split (3 seeds): GELU 0.20, SiLU 0.21, Tanh 0.38, ReLU 0.57 GHz. Smooth activations (SiLU, GELU) are the most accurate here; SiLU is as good as the best and cheaper than GELU.

Ensemble size

Measured on the validation split: 1 network 0.19, 2 networks 0.18, 3 networks 0.17, 5 networks 0.17, 10 networks 0.17 GHz. Averaging cancels part of each network's random error: 5 networks are 13 % more accurate than one, and going to 10 doubles the cost for almost nothing (under 0.01 GHz). The spread between the 5 also serves as an uncertainty estimate.

Does the spread mean anything?

Where the 5 networks disagree, the prediction is more likely to be wrong: on the validation cells the 10 % with the largest spread have an average error of 0.49 GHz against 0.13 GHz for the rest (correlation 0.63). So the spread flags risky predictions; it is shown in the app as the ensemble spread.

Training length

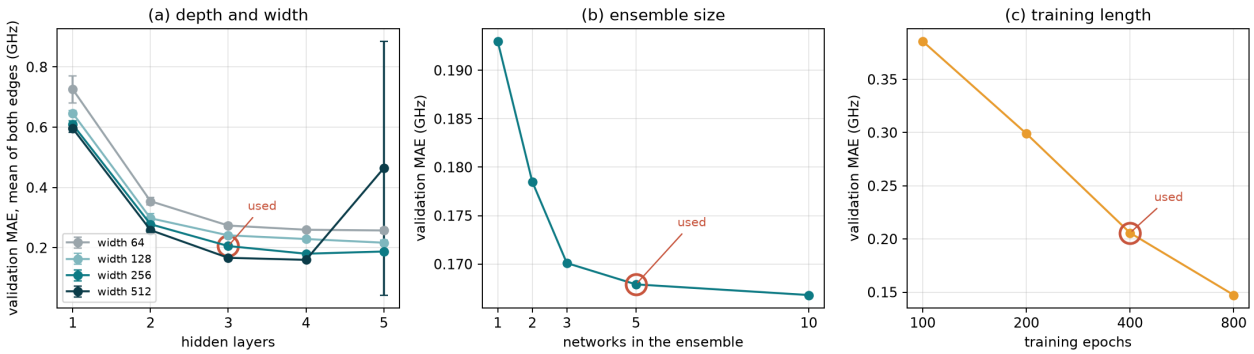
Validation error by training length: 100 epochs 0.39, 200 epochs 0.30, 400 epochs 0.21, 800 epochs 0.15 GHz (about 30 s vs 60 s per network for 400 and 800). So 400 epochs was not fully converged: 800 epochs is 0.06 GHz better (28 %). 400 was a time trade-off (the whole pipeline is retrained after every new COMSOL export). Like the network size, this changes no conclusion: the forward error is already far below the 1 GHz criterion and below the PWE-COMSOL difference. Training longer is an easy improvement for Semester IV.

Tandem fabrication penalty

The penalty pushes designs away from thin walls and tiny radii while training. Tested on 200 validation targets (designs checked with the exact rules, edges by the forward network): $\lambda = 0$: 90.5 % of designs fabricable, 90.5 % fabricable and within 1 GHz; $\lambda = 0.001$: 100 % of designs fabricable, 100 % fabricable and within 1 GHz; $\lambda = 0.01$: 100 % of designs fabricable, 100 % fabricable and within 1 GHz. $\lambda = 0.001$ and 0.01 are equally good on this measure; without the penalty only 90.5 % of designs can be fabricated. The smaller weight disturbs the band-edge accuracy least (worst-edge error 0.041 GHz vs 0.044), so 0.001 was kept.

CVAE latent size and beta

Tested on 100 validation targets (64 samples each, before refining; scored by the forward network and the exact fabrication rules). Latent size: 2 numbers: 4.2 distinct designs within 1 GHz (of 8), 8 % of samples within 1 GHz; 4 numbers: 4.6 distinct designs within 1 GHz (of 8), 8 % of samples within 1 GHz; 8 numbers: 6.9 distinct designs within 1 GHz (of 8), 22 % of samples within 1 GHz. β: 0.1: 3.7 designs, 6 % of samples; 0.5: 4.6 designs, 8 % of samples; 1: 4.9 designs, 9 % of samples. A larger code helps: latent size 8 gives 6.9 designs against 4.6 for the 4 used. That fits the geometry of the problem: with the 2 band edges fixed, the 10 design numbers leave 8 free directions, and a 4-number code cannot cover them all. 4 was a standard small choice made early; the refine step makes up for most of the difference (CVAE + refine already gives about 8 designs per target at 97 %), so a larger latent code is an easy upgrade for Semester IV. For β, 1 came out best (4.9 designs vs 4.6 at the 0.5 used): β trades how closely the codes follow the normal distribution (needed so that random codes decode to valid designs) against how much detail they keep.



Validation error (mean of both edges) of single networks: (a) by depth and width, (b) by number of networks averaged, (c) by training length. Error bars: spread over 3 seeds.

All depth / width results

Hidden layers	Width	Weights	Validation MAE f_{lower} / f_{upper} (GHz)	Mean	Seed spread
1	64	1,346	0.700 / 0.751	0.726	0.045
1	128	2,690	0.608 / 0.685	0.646	0.008
1	256	5,378	0.582 / 0.636	0.609	0.013
1	512	10,754	0.586 / 0.606	0.596	0.013
2	64	5,506	0.284 / 0.424	0.354	0.012
2	128	19,202	0.230 / 0.366	0.298	0.015
2	256	71,170	0.213 / 0.343	0.278	0.007
2	512	273,410	0.211 / 0.306	0.259	0.003
3	64	9,666	0.225 / 0.322	0.274	0.007
3	128	35,714	0.194 / 0.288	0.241	0.006
3	256	136,962	0.175 / 0.237	0.206	0.001
3	512	536,066	0.144 / 0.189	0.167	0.002
4	64	13,826	0.218 / 0.301	0.260	0.004
4	128	52,226	0.194 / 0.264	0.229	0.006
4	256	202,754	0.164 / 0.197	0.180	0.001
4	512	798,722	0.143 / 0.176	0.160	0.003
5	64	17,986	0.224 / 0.291	0.257	0.008
5	128	68,738	0.191 / 0.243	0.217	0.008
5	256	268,546	0.167 / 0.208	0.188	0.005
5	512	1,061,378	0.368 / 0.560	0.464	0.421